

Theory of Trees and Forests: A Mathematical Framework for Quantum Gravity, Particle Physics, and Cosmology

Macharia Bari

bariimacharia@gmail.com

July 26, 2026

Preprint: TOF-2026-07-26

Abstract

We present the **Theory of Trees and Forests (TOF)**—a mathematical framework in which the fundamental degrees of freedom are discrete “trees” (nodes of a spin network) embedded in a Lorentz-invariant causal set, connected by “roots” (gauge fields) and “soil” (scalar fields). The framework:

1. **Calibrates** its six fundamental parameters to six measured anchors through dimensionally consistent relations, explicitly separating definitions from predictions.
2. **Derives** the hydrodynamic reduction of the Bose–Hubbard tree lattice to an analogue curved metric, and the standard Klein–Gordon to Schrödinger reduction.
3. **Postulates** the Standard Model gauge and matter content as a one-to-one root-network/soliton structure.
4. **Separates established results from conjectures:** the emergent Einstein dynamics, asymptotic safety, and the sparticle spectrum are stated as open questions, not results.

The framework is presented as a well-posed, internally consistent research proposal with explicit open problems (the hierarchy problem and the cosmological-constant problem are stated rather than hidden).

1 Introduction

1.1 Motivation

The unification of quantum mechanics and general relativity remains the foremost challenge in theoretical physics. Existing approaches—string theory, loop quantum gravity, causal sets, and asymptotic safety—each capture fragments of the full picture but lack a unifying physical intuition. In this work we propose that the fundamental degrees of freedom are discrete “trees” (nodes of a spin network) connected by “roots” (gauge fields) and embedded in a “soil” (scalar field) that gives mass and drives cosmic expansion.

1.2 The Core Idea

The Theory of Trees and Forests (TOF) replaces the smooth manifold of general relativity with a **causal set** of trees. Each tree is a quantum harmonic oscillator (sway mode) coupled to its neighbors via root grafts (gauge connections). The collective behavior of these trees is conjectured to produce:

- **Emergent spacetime:** an effective Lorentzian metric arising in the hydrodynamic limit of the tree lattice (Section 5).
- **Quantum fields:** tree excitations mapped one-to-one onto the particles of the Standard Model (Section 4).
- **Gravity:** curvature of the forest as the gravitational field.

1.3 Scope and Status

We distinguish explicitly three kinds of statements used in this paper:

Postulate. A defining assumption of the framework, not derived (Section 2).

Derived. A statement obtained from the postulates by explicit, shown steps (the hydrodynamic reduction of Section 5 and the non-relativistic limit of Section 6).

Conjecture. A claim motivated but not established by the framework, flagged as such (asymptotic safety, emergence of the Einstein–Hilbert dynamics, the analogue of the Standard Model sector beyond the kinematical level).

Borrowed formalisms (causal sets, spin networks, Bose–Hubbard lattice, Schwinger–Keldysh contour, group field theory) are used as *tools*; they are not claimed to be identical structures unless a precise identification is given.

2 The Fundamental Substrate

2.1 The Causal Set Forest

Let $\mathcal{C}$ be a locally finite, partially ordered set (poset) of events x_i with order relation $x_i \prec x_j$ (meaning event i causally precedes event j). Each event x_i is a “tree” in the forest.

Properties:

- **Local finiteness:** Each event has finitely many ancestors and descendants
- **Transitivity:** $x \prec y$ and $y \prec z \Rightarrow x \prec z$
- **Acyclicity:** No closed causal loops

The **spacetime volume** between events is taken to be the volume of the causal interval (a causal diamond) in 1+1 dimensions,

$$V(x_i, x_j) = \frac{\pi}{24}(\Delta t^2 - \Delta \mathbf{x}^2)^2, \quad (1)$$

in units $c = 1$ (Bombelli et al. [1]). This is a standard causal-set result for the volume of a causal interval in flat spacetime, not an arbitrary ansatz.

Remark 2.1. *We emphasize that Eq. (1) is the standard Minkowski-space causal-interval volume and is used here only in that context.*

2.2 The Tree Hilbert Space

Each tree i is a quantum harmonic oscillator with:

- **Sway mode:** $\hat{x}_i$ (position operator)
- **Momentum:** $\hat{p}_i$ (mass flow operator)
- **Occupation number:** $\hat{n}_i = \hat{b}_i^\dagger \hat{b}_i$ (sway quanta)

The **single-tree Hamiltonian** is a Bose–Hubbard on-site term,

$$\hat{h}_i = \hbar\omega_i \left(\hat{n}_i + \frac{1}{2} \right) + \frac{U_i}{2} \hat{n}_i(\hat{n}_i - 1) \quad (2)$$

where $\omega_i = c_{\text{TOF}}/H_i$ with c_{TOF} the internal (sound) speed c_s defined in Eq. (20) below, H_i the local height, and

$$U_i = \frac{\hbar^2}{2m_i a_i^2}, \quad (3)$$

the standard on-site repulsion of a lattice of mass m_i on spacing a_i . The frequency ω_i could in principle carry the factor 4.493 (the first non-zero root of $\tan x = x$, appropriate for a specific vibrating-string boundary condition); here the bare value c/H is used.

2.3 The Full Hilbert Space

The total Hilbert space is the tensor product over all trees:

$$\mathcal{H} = \bigotimes_{i \in \mathcal{C}} \mathcal{H}_i. \quad (4)$$

A general state is:

$$|\Psi\rangle = \sum_{\{n_i\}} \Psi(\{n_i\}) |\{n_i\}\rangle. \quad (5)$$

where $\{n_i\}$ is the occupation-number configuration of the entire forest.

3 The Master Lagrangian

The full action of the TOF is assembled from the following terms. For each term we state its status:

$$S_{\text{TOF}} = S_{\text{causal}} + S_{\text{BH}} + S_{\text{YM}} + S_{\text{fermion}} + S_{\text{Higgs}} + S_{\text{gravity}} + S_{\text{SUSY}} + S_{\text{axion}} + S_{\text{SK}}. \quad (6)$$

Causal Set Action (Postulate):

$$S_{\text{causal}} = \frac{1}{\hbar} \sum_{x_i \prec x_j} \left(m^2 V(x_i, x_j) - \frac{\kappa}{2} \ln \left(\frac{\Delta x^2}{a^2} \right) \right). \quad (7)$$

Bose–Hubbard Action (Postulate; the microscopic substrate):

$$S_{BH} = \int d\tau \left[\sum_i \bar{b}_i \frac{db_i}{d\tau} - \sum_{\langle i,j \rangle} J_{ij} (\bar{b}_i b_j + \text{h.c.}) - \sum_i \frac{U_i}{2} \bar{b}_i^2 b_i^2 \right]. \quad (8)$$

Yang–Mills Action (Root gauge fields; Postulate):

$$S_{YM} = -\frac{1}{4} \int d^4x \sqrt{-g} \text{Tr}(F_{\mu\nu} F^{\mu\nu}), \quad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + ig[A_\mu, A_\nu], \quad (9)$$

with $A_\mu \in \mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$.

Fermion Action (Root knots; Postulate):

$$S_{fermion} = \int d^4x \sqrt{-g} \sum_f \bar{\psi}_f (i\gamma^\mu D_\mu - m_f) \psi_f. \quad (10)$$

Higgs Action (Soil moisture; Postulate):

$$S_{Higgs} = \int d^4x \sqrt{-g} (|D_\mu \phi|^2 - V(\phi)), \quad V(\phi) = -\frac{1}{2} \mu^2 |\phi|^2 + \frac{\lambda}{4!} |\phi|^4. \quad (11)$$

Gravity Action (Emergent metric; *Conjecture* G_1 , see Section 5):

$$S_{gravity} = \frac{1}{16\pi G_{eff}} \int d^4x \sqrt{-g} (R - 2\Lambda_{eff}). \quad (12)$$

Supersymmetry Action (*optional, conjectural* extension):

$$S_{SUSY} = \int d^4x \sqrt{-g} \left(\partial_\mu \phi^* \partial^\mu \phi + i\bar{\psi} \not{\partial} \psi + |F|^2 + \left(\frac{\partial W}{\partial \phi} \psi \psi + \text{h.c.} \right) \right). \quad (13)$$

Axion Action (Soil/“dark energy” candidate; Postulate):

$$S_{axion} = \int d^4x \sqrt{-g} \left(\frac{1}{2} \partial_\mu a \partial^\mu a - V(a) - \frac{a}{f_a} G\tilde{G} \right), \quad V(a) = \Lambda_{QCD}^4 \left(1 - \cos \left(\frac{a}{f_a} \right) \right). \quad (14)$$

Schwinger–Keldysh Action (Unitarity and dissipation; a *tool* for real-time correlation functions):

$$S_{SK} = \int_{\text{contour}} d^4x \sqrt{-g} \mathcal{L}_{matter}[\phi_+, J_+] - \mathcal{L}_{matter}[\phi_-, J_-]. \quad (15)$$

Remark 3.1. “Root interactions” are the Bose–Hubbard hopping (J) and on-site (U) interactions; there are no additional undefined interaction terms.

4 Emergence of the Standard Model

4.1 Gauge Symmetries from Root Networks

The root network of the forest is modeled as a **principal G -bundle** over the causal set, with

$$G = SU(3) \times SU(2) \times U(1). \quad (16)$$

This is a *postulate* (an input assignment of gauge structure), not a derived result. The gauge bosons are the connection modes of the root network; they never arise from single-tree modes.

Group	Root Type	Mediator Boson	Coupling (calibrated)
$SU(3)$	Deep root grafts	Gluons (g)	$g_s = \sqrt{4\pi\alpha_s}$
$SU(2)$	Shallow root grafts	$W^\pm, Z$	g_w
$U(1)$	Bark sheath	Photon (γ)	e

Table 1: Gauge symmetries from root networks (postulate).

4.2 Particle Content from Tree Excitations

The mapping between excitations and Standard-Model particles is made *one-to-one*: each physical particle corresponds to exactly one well-defined excitation mode. Gauge bosons are connection modes of the root network; matter fermions are solitonic modes; the Higgs is the fluctuation of the soil condensate.

Tree Mode	Excitation	Mass	Spin	Charge	Sector
Root connection, $U(1)$	Photon (γ)	0	1	0	gauge
Root connection, $SU(3)$	Gluon (g)	0	1	color	gauge
Root connection, $SU(2)$	$W^\pm, Z$	m_W, m_Z	1	$\pm 1, 0$	gauge
Soil fluctuation	Higgs (h)	$m_h = \sqrt{2\lambda} v$	0	0	scalar
Bark soliton	Electron (e^-)	$m_e = \frac{\hbar^2}{2a^2 E_{bark} y_q \bar{v}}$	1/2	-1	lepton
Deep-root soliton	Quark (q)	$m_q = \frac{\sqrt{2}}{\hbar}$	1/2	$2/3, -1/3$	quark
Soil pore mode	Neutrino (ν)	$m_\nu = \frac{\hbar}{c L_{root}}$	1/2	0	lepton

Table 2: Particle content from tree excitations (one-to-one mapping).

Remark 4.1. *Three mass formulas deserve comment.*

1. *Quark masses are routed through the Yukawa matrix, as in the Standard Model; they are time-independent, unlike a naive tree-geometry formula.*
2. *The proton is a composite state, $m_p = \sum_q m_{q,val} + E_{bind}/c^2 \approx 938.3 \text{ MeV}$; it is not given by a single-particle formula.*
3. *$m_e = \hbar^2/(2a^2 E_{bark})$ is dimensionally consistent only if E_{bark} is an energy (J). With $m_e = 9.109 \times 10^{-31} \text{ kg}$, $E_{bark} \approx 2.3 \times 10^{31} \text{ J}$.*

4.3 The Yukawa Sector

The Higgs field (soil) couples to fermions (root knots) via

$$\mathcal{L}_{Yukawa} = - \sum_{i,j} y_{ij} \bar{\psi}_i \phi \psi_j. \quad (17)$$

The **Yukawa matrix** y_{ij} is the “root graft mixing matrix.” It is a free input of the framework; TOF does not yet predict the observed fermion masses and mixings. Reproducing the observed hierarchy of Yukawa couplings would require additional structure; this is an open problem.

5 Emergence of General Relativity

5.1 From Bose–Hubbard to a Curved Effective Metric

We now show the steps by which a Bose–Hubbard lattice reproduces relativistic propagation on an emergent curved metric. This is the standard “analogue gravity” construction (Unruh [3]; see [4] for a review), and it is the only part of the gravity sector that is *derived* here.

Step 1 (mean field). Take the continuum limit of the Bose–Hubbard model, $a \rightarrow 0$, $b_i \rightarrow \sqrt{a^d} \varphi(\mathbf{x})$, $J \rightarrow \hbar^2/(2ma^2)$, $U \rightarrow \lambda$. In a coherent state the equation of motion is the discrete nonlinear Schrödinger (Gross–Pitaevskii) equation,

$$i\hbar \frac{\partial \varphi}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \varphi + \lambda |\varphi|^2 \varphi - \mu \varphi. \quad (18)$$

Step 2 (hydrodynamics). Write $\varphi = \sqrt{\rho} e^{i\theta}$ (Madelung transform) and define the superfluid velocity $\mathbf{v} = (\hbar/m) \nabla \theta$. Eq. (18) becomes the continuity equation and an Euler equation with a quantum-pressure term,

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{v}) = 0, \quad m \frac{\partial \mathbf{v}}{\partial t} = -\nabla \left(\mu + \lambda \rho + \frac{\hbar^2}{2m^2} \frac{\nabla^2 \sqrt{\rho}}{\sqrt{\rho}} \right). \quad (19)$$

Step 3 (phonons). Linearizing Eq. (19) about a stationary background $(\rho_0, \mathbf{v}_0)$ yields sound waves with speed

$$c_s = \sqrt{\frac{\lambda \rho_0}{m}}, \quad (20)$$

which is the internal (“TOF”) speed. The causal set’s volume scale $\ln(H/a)$ enters through the identification $c = c_s \ln(H/a)$ (Section 7).

Step 4 (emergent metric). Low-energy fluctuations of the condensate do not propagate on the flat background; they propagate on the effective (acoustic) metric

$$ds^2 = \frac{\rho_0}{c_s} \left[- (c_s^2 - v_0^2) dt^2 - 2v_{0i} dx^i dt + \delta_{ij} dx^i dx^j \right], \quad (21)$$

(Unruh [3]). This is the concrete sense in which “spacetime is emergent”: a curved Lorentzian metric emerges from the collective motion of the substrate, and quasiparticles (sway quanta) follow geodesics of Eq. (21).

5.2 The Einstein–Hilbert Action: Conjecture G_1

Conjecture 5.1 (Emergent Einstein–Hilbert action). *After integrating out the substrate at long wavelengths, the effective action for the metric (21) takes the form*

$$\Gamma[g] = \frac{c^4}{16\pi G_{eff}} \int d^4x \sqrt{-g} R + \text{higher-order curvature terms}, \quad (22)$$

with

$$G_{eff} = \frac{c^3 a^2}{\hbar} = G_{\text{Newton}}, \quad (23)$$

where the last equality holds because the tree spacing is calibrated to the Planck length, $a = l_P$ (Section 7).

Remark 5.2. *Conjecture 5.1 is the dynamical content of emergent gravity and is not derived in this paper. The kinematic part (Eq. (21)) is established in the analogue gravity literature; the dynamics (that the metric itself obeys Einstein's equations sourced by the matter stress tensor) requires computing the self-interaction of the emergent metric, which is an open problem.*

5.3 Black Hole Solutions

Assuming Conjecture 5.1, the Schwarzschild metric is

$$ds^2 = -c^2 \left(1 - \frac{2G_{eff}M}{c^2 r}\right) dt^2 + \left(1 - \frac{2G_{eff}M}{c^2 r}\right)^{-1} dr^2 + r^2 d\Omega^2, \quad (24)$$

with Schwarzschild radius

$$r_s = \frac{2G_{eff}M}{c^2}, \quad (25)$$

and Hawking temperature

$$T_H = \frac{\hbar c^3}{8\pi k_B G_{eff}M}, \quad (26)$$

both standard general-relativistic results that follow from the emergent metric Eq. (21) if G_1 holds.

6 Emergence of Quantum Mechanics

6.1 From the Klein–Gordon Equation

On the emergent metric, a free scalar obeys the Klein–Gordon equation

$$\frac{1}{\sqrt{-g}} \partial_\mu (\sqrt{-g} g^{\mu\nu} \partial_\nu \phi) = 0. \quad (27)$$

6.2 Non-Relativistic Limit

Assume $\phi(x, t) = \psi(x, t)e^{-imc^2 t/\hbar}$. Substituting and keeping terms to order $1/c^2$ gives, in the Minkowski background of the emergent metric,

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \psi + V(x)\psi. \quad (28)$$

This is the standard derivation (not a new result); its TOF content is only that the metric on which the reduction is performed is the emergent acoustic metric of Section 5.

6.3 The Ehrenfest Theorem

Taking expectation values of the position and momentum operators built from Eq. (28),

$$\frac{d\langle \hat{x} \rangle}{dt} = \frac{\langle \hat{p} \rangle}{m}, \quad \frac{d\langle \hat{p} \rangle}{dt} = -\left\langle \frac{\partial V}{\partial x} \right\rangle, \quad (29)$$

which in the classical limit is Newton's second law,

$$\frac{d^2 \langle \hat{x} \rangle}{dt^2} = -\frac{1}{m} \nabla V(\langle x \rangle). \quad (30)$$

This is standard; it is included only to close the chain “quantum fields $\rightarrow$ quantum mechanics $\rightarrow$ classical mechanics”.

7 The Constants: Calibration, Not Derivation

This section presents the “Full Constant Table,” the fundamental parameters, and their numerical values. The six parameters are *calibrated* to measured anchors through dimensionally consistent relations; the procedure is a well-posed fit, not a derivation of the constants from nothing.

7.1 The Six Fundamental Parameters

The TOF has six free parameters:

1. a = tree spacing (length),
2. H = forest radius (length),
3. ρ = tree mass density (kg m^{-3}),
4. E = elastic energy density (J m^{-3}),
5. J = root graft stiffness (energy, J),
6. U = on-site repulsion (energy, J).

7.2 Dimensionally Consistent Defining Relations

The following relations are dimensionally consistent and are the *definitions* used below (the subscripts on λ and κ indicate dimensionless numbers):

$$c = c_s \ln\left(\frac{H}{a}\right), \quad c_s \equiv \sqrt{\frac{E}{\rho}}, \quad (31)$$

$$\hbar = \rho a^4 c_s, \quad \text{from } m_t a^2 \omega, \ m_t = \rho a^3, \ \omega = c_s/a, \quad (32)$$

$$a = l_P = \sqrt{\frac{\hbar G}{c^3}}, \quad \text{equivalently } G = \frac{c^3 a^2}{\hbar}, \quad (33)$$

$$H = \frac{c}{H_0}, \quad (34)$$

$$\Lambda_{eff} = \frac{\bar{\lambda}}{H^2}, \quad \bar{\lambda} \equiv \Lambda H^2, \quad (35)$$

$$m_h = \kappa \frac{\hbar}{cH}, \quad \kappa \equiv \frac{m_h c H}{\hbar}. \quad (36)$$

Remark 7.1. Dimensional consistency of the defining relations.

- $\hbar = a^2 \sqrt{\rho E} \ln(H/a)$ would have units kg/s , not J s ; the consistent form is Eq. (32).

- $G = \frac{c^4}{16\pi\rho\ln(H/a)} \frac{1}{1+J/U}$ would have units $\text{m}^7/(\text{kg s}^4)$, not $\text{m}^3/(\text{kg s}^2)$; the consistent form is Eq. (33).
- $\Lambda = -\frac{1}{2}\xi(a/H)^2$ would be dimensionless rather than m^{-2} ; the consistent form is Eq. (35).
- A naive formula $m_q = \frac{U}{2} \frac{a}{H} e^{-\gamma t}$ would have units of energy (J), vary in time, and conflate distinct particles; quark masses are instead routed through the Yukawa matrix (Section 4).

7.3 Calibration of the Parameters

Eqs. (31)–(36) are inverted for the six parameters, using the measured values of c , $\hbar$, G , H_0 , Λ , and m_h . The gauge-sector parameters J and U are fixed by the strong coupling through the relation $\alpha_s = J^2/(\hbar c U)$ with $g_s = \sqrt{4\pi\alpha_s}$:

Parameter	Relation	Calibrated value	Meaning
a	Eq. (33)	$1.616 \times 10^{-35} \text{ m}$	tree spacing = l_P
H	Eq. (34)	$1.372 \times 10^{26} \text{ m}$	forest radius = c/H_0
ρ	Eq. (32)	$7.2 \times 10^{98} \text{ kg/m}^3$	tree density
E	Eq. (31)	$3.3 \times 10^{111} \text{ J/m}^3$	elastic energy density
$J = g_s \hbar c$	$\alpha_s(m_Z) = 0.118$	$3.85 \times 10^{-26} \text{ J}$	root graft stiffness
$U = J^2/(\hbar c \alpha_s)$	$\alpha_s(m_Z) = 0.118$	$3.97 \times 10^{-25} \text{ J}$	on-site repulsion
$c_s = c/\ln(H/a)$	Eq. (31)	$2.14 \times 10^6 \text{ m/s}$	internal (sound) speed
$\bar{\lambda} = \Lambda H^2$	Eq. (35)	2.05	dimensionless CC
$\kappa = m_h c H/\hbar$	Eq. (36)	8.7×10^{43}	dimensionless Higgs ratio

Table 3: Calibrated tree parameters.

7.4 What Is Derived and What Is Not

It must be stressed that Eqs. (31)–(36) are *definitions*: they reproduce c , $\hbar$, G , and H_0 exactly by construction. They are not predictions. The genuinely nontrivial dimensionless numbers are $\bar{\lambda}$ and κ :

- $\bar{\lambda} = \Lambda H^2 \approx 2.05$ is *not* predicted by the framework; it encodes the cosmological-constant problem.
- $\kappa \approx 8.7 \times 10^{43}$ encodes the *hierarchy problem*: the natural mass scale of a theory with $\hbar, c, a$ is the Planck scale, and reproducing $m_h = 125.1 \text{ GeV}$ requires a dimensionless ratio of order 10^{17} in the relation $m_h = \kappa \hbar/(cH)$. TOF does not solve either problem; they remain open.

7.5 Particle Masses

Fermion masses are routed through the Yukawa matrix (Section 4),

$$m_f = \frac{y_f v}{\sqrt{2}}, \quad v = 246.2 \text{ GeV}, \quad (37)$$

with y_f free inputs. The neutrino mass $m_\nu = \hbar/(c L_{root})$ is dimensionally consistent; the bound $m_\nu \lesssim 0.12$ eV implies a pore scale $L_{root} \gtrsim 1.6 \mu\text{m}$. The proton is composite:

$$m_p = \sum_q m_{q,\text{val}} + \frac{E_{\text{bind}}}{c^2} \approx 938.3 \text{ MeV}. \quad (38)$$

8 Experimental Predictions

The following tables list consequences of the framework. They are *model-dependent targets*: none of the numbers is a sharp, derived prediction of the framework as it stands.

8.1 Quantum Gravity Regime

Item	Observable	Status
Emergent metric	Phonon/graviton kinematics	Derived (Sec. 5)
Einstein dynamics	Curvature response	Conjecture G_1
Area discreteness	Area spectra $A \sim l_P^2 \sqrt{j(j+1)}$	Model-dependent
Lorentz invariance	Photon dispersion $\Delta c/c \sim (E/M_P)^2$ (upper bound)	Open question
Graviton mass	$m_g = 0$ at tree level (no anomalous GW dispersion)	Consistent w/ Lorentz inv
Black hole entropy	$S_{BH} = A/(4l_P^2) + \alpha \ln(A)$	Standard result

Table 4: Quantum-gravity consequences.

Remark 8.1. *A massive graviton would break the Lorentz invariance of the causal set (Conjecture 12.7); at tree level the graviton is therefore massless. Any Lorentz violation would appear only as $\Delta c/c \sim (E/M_P)^2$ corrections, bounded by GLAST/Fermi.*

8.2 Particle Physics Regime

Item	Observable	Status	Experimental
Higgs couplings	SM-like; deviations would probe root-network structure	Conjectural	HL-LHC
Supersymmetry	Sparticles $m_{\tilde{q}} \sim 1\text{--}10$ TeV (if SUSY realized)	Conjectural	Future colliders
Neutrino seesaw	$m_R \sim 10^{14}$ GeV	Conjectural	Neutrino experiments
Axion dark matter	m_a model-dependent ($\sim 10^{-22}$ eV if ultralight)	Conjectural	Axion experiments

Table 5: Particle-physics targets.

8.3 Cosmology Regime

9 The Renormalization Group Flow

The beta functions of the framework are *not computed* in this paper. The fixed-point table below is illustrative; establishing it requires a functional renormalization group computation.

Item	Observable	Status	Experiment
Inflation	Primordial tensor modes $r \sim 0.001\text{--}0.01$	Model-dependent	CMB-S4 / LiteB
Dark energy	Equation of state $w(a) = -1 + w_0(1 - a)$	Model-dependent	Euclid / DES
Cosmic structure	Non-Gaussianity $f_{NL} \sim \mathcal{O}(1)$	Model-dependent	LSS surveys
CMB power spectrum	Damping tail $\ell_{\text{damping}} \sim 3000$	Standard	Planck / CMB-S4

Table 6: Cosmology targets.

Conjecture 9.1 (Asymptotic safety). *The theory possesses a non-Gaussian fixed point in the UV. This is a conjecture modeled on asymptotically safe quantum gravity (Weinberg [8], Reuter [9]); establishing it for the TOF requires a functional renormalization group computation of the beta functions of Eq. (39), which has not been performed.*

The beta functions, in the usual perturbative form, would read

$$\beta_{g_i} = \mu \frac{dg_i}{d\mu} = (d_i - 4)g_i + b_i g_i^2 + c_i g_i^3 + \dots \quad (39)$$

Coupling	Fixed Point	Value (illustrative)	UV Behavior
G_{eff}	G_*	$\sim 10^{-2} M_P^{-2}$	Asymptotically safe
λ (Higgs)	λ_*	0.1	Stabilized
y_t (top Yukawa)	y_*	0.8	Marginal
g_s (strong)	g_*	1.2	Freezes
g_w (weak)	g_*	0.65	Freezes
e (EM)	e_*	0.3	Landau pole avoided

Table 7: **Illustrative only.** No computation of these numbers has been performed; they are placeholders consistent with Conjecture 9.1.

Remark 9.2. *The relation $\mu_{GUT} = (J/U)\hbar c/a^2$ is dimensionally an energy but numerically evaluates to $\sim 10^{54}$ GeV, not 10^{16} GeV, so it cannot serve as a unification scale. Any gauge-coupling unification below the Planck scale is an open, uncomputed question; the natural scale of the framework is Planckian.*

10 The Problem of Time and Observer

10.1 The Wheeler–DeWitt Equation

The Hamiltonian constraint for the forest universe (assuming Conjecture 5.1),

$$\hat{\mathcal{H}}|\Psi\rangle = 0, \quad \hat{\mathcal{H}} = \hat{\mathcal{H}}_{gravity} + \hat{\mathcal{H}}_{matter}, \quad (40)$$

is the standard Wheeler–DeWitt equation. It stands as a consistency condition, not a solved problem; in particular, its resolution is an open issue of quantum gravity generally.

10.2 Relational Observables

Define observables as functions of the matter fields:

$$\mathcal{O}[\Phi] = \int d^4x \sqrt{-g} \mathcal{F}[\Phi(x)], \quad (41)$$

with inner product

$$\langle \Psi | \mathcal{O} | \Psi \rangle = \int \mathcal{D}[\Phi] \Psi^*[\Phi] \mathcal{O}[\Phi] \Psi[\Phi]. \quad (42)$$

10.3 The Causal Set Clock

Time emerges from the partial order of the causal set:

$$\tau(x) = \sum_{y \prec x} (\text{volume of causal diamond}), \quad (43)$$

using Eq. (1). This is the internal clock of the forest.

11 The Information Paradox Resolution

11.1 Entanglement Entropy of the Forest

For a region R of the forest,

$$S_{ent}(R) = -\text{Tr}(\hat{\rho}_R \log_2 \hat{\rho}_R), \quad \hat{\rho}_R = \text{Tr}_{\bar{R}} |\Psi\rangle \langle \Psi|. \quad (44)$$

11.2 The Page Curve

For a black hole (fallen forest patch) of age t ,

$$S_{BH}(t) = \min \left(\frac{A}{4l_P^2}, \frac{A}{4l_P^2} \left(1 - \frac{t}{t_{evap}} \right) \right), \quad (45)$$

with evaporation time

$$t_{evap} = \frac{5120\pi G_{eff}^2 M^3}{\hbar c^4}. \quad (46)$$

Information is preserved via the entanglement between the black hole and the outside forest (Page [11]):

$$\frac{dS_{BH}}{dt} = -\frac{dS_{outside}}{dt} + \mathcal{O}(e^{-S_{BH}}). \quad (47)$$

This is the standard Page-curve argument (Page [11]), applied here to the forest geometry.

12 Consistency: Established Results and Conjectures

We separate what is established (standard results invoked by the framework) from what is conjectural.

12.1 Established (standard results)

Theorem 12.1 (Gauge symmetry). *The root network action is invariant under local G -transformations:*

$$A_\mu \rightarrow U A_\mu U^{-1} - \frac{i}{g} U \partial_\mu U^{-1}, \quad (48)$$

because $F_{\mu\nu}$ transforms covariantly, $F_{\mu\nu} \rightarrow U F_{\mu\nu} U^{-1}$, so $\text{Tr}(F_{\mu\nu} F^{\mu\nu})$ is invariant.

Theorem 12.2 (Unitarity of the Lindblad form). *The generator*

$$\frac{d\hat{\rho}}{dt} = -\frac{i}{\hbar}[\hat{H}, \hat{\rho}] + \sum_i \gamma_i \left(\hat{L}_i \hat{\rho} \hat{L}_i^\dagger - \frac{1}{2} \{ \hat{L}_i^\dagger \hat{L}_i, \hat{\rho} \} \right) \quad (49)$$

preserves the trace and positivity of $\hat{\rho}$ (Lindblad [10]). This is a theorem about the form, not a claim that the TOF dynamics is unitary; whether the emergent dynamics is of this form with a valid dissipator $\mathcal{D}[\hat{\rho}]$ is an open question.

Theorem 12.3 (CPT symmetry of the SK contour). *The Schwinger–Keldysh contour is symmetric under CPT: $\Theta \mathcal{H} \Theta^{-1} = \mathcal{H}$. This is a standard property of the closed-time-path formalism.*

Theorem 12.4 (On-shell Wess–Zumino SUSY invariance). *The Wess–Zumino action is invariant under*

$$\delta\phi = \sqrt{2}\epsilon\psi, \quad \delta\psi = i\sqrt{2}\sigma^\mu\bar{\epsilon}\partial_\mu\phi. \quad (50)$$

This is the standard result of supersymmetry; it is a property of the optional SUSY extension, not a derivation that the TOF is supersymmetric.

Theorem 12.5 (Equivalence principle (definitional)). *All matter fields couple minimally to the same emergent metric, so*

$$\frac{\delta S_{\text{matter}}}{\delta g_{\mu\nu}} = \frac{1}{2}\sqrt{-g}T^{\mu\nu}. \quad (51)$$

This is the definition of minimal coupling, not a derived physical law.

12.2 Conjectural

Conjecture 12.6 (Renormalizability / asymptotic safety). *The full theory is renormalizable via a non-Gaussian fixed point. See Conjecture 9.1; no beta functions have been computed.*

Conjecture 12.7 (Lorentz invariance of the causal set). *The causal-set discretization preserves local Lorentz symmetry, in the sense that a random “sprinkling” of trees in Minkowski space has no preferred frame:*

$$P(x, y) = P(\Lambda x, \Lambda y), \quad \Lambda \in SO(3, 1). \quad (52)$$

This is the standard causal-set argument; note that it establishes invariance of the ensemble of sprinklings, not of any single configuration. Whether the observed single universe experiences residual effects at $O(E/M_P)^2$ is an open question (see Section 8).

Conjecture 12.8 (Emergent Einstein dynamics). *Conjecture 5.1: the metric Eq. (21) obeys Einstein’s equations.*

13 The Cosmic Forest

13.1 The Friedmann Equations

Assuming the emergent Einstein equations (Conjecture 5.1), the forest density $\rho_{tree}(t)$ obeys the standard Friedmann system,

$$\dot{\rho}_{tree} + 3H_{forest}(\rho_{tree} + p_{tree}) = 0, \quad H_{forest} = \frac{\dot{R}_{forest}}{R_{forest}}, \quad (53)$$

$$H_{forest}^2 = \frac{8\pi G_{eff}}{3}\rho_{tree} - \frac{k}{R_{forest}^2} + \frac{\Lambda_{eff}}{3}. \quad (54)$$

13.2 Inflation

Inflation occurs when $\ddot{R}_{forest} > 0$. A proposed inflaton is the tree height field $H(t)$, with

$$V(H) = \frac{\lambda}{4}H^4 \ln^2(H/a), \quad (55)$$

and the standard slow-roll parameters

$$\epsilon = \frac{1}{2} \left(\frac{V'}{V} \right)^2 \ll 1, \quad \eta = \frac{V''}{V} \ll 1. \quad (56)$$

This potential is a *model* (a choice), not a derived result.

13.3 Dark Energy

The soil field $\phi(t)$ drives late-time acceleration,

$$\rho_{DE} = \frac{1}{2}\dot{\phi}^2 + V(\phi), \quad p_{DE} = \frac{1}{2}\dot{\phi}^2 - V(\phi), \quad (57)$$

with equation of state

$$w_{DE} = \frac{p_{DE}}{\rho_{DE}} = -1 + \frac{\dot{\phi}^2}{V(\phi)}. \quad (58)$$

Whether the observed $w \simeq -1$ is accounted for by the axion potential $V(a)$ without severe fine-tuning is the cosmological-constant problem, which the framework does not resolve (Section 7).

14 Conclusions

We have presented the Theory of Trees and Forests. The framework:

1. **Postulates** a causal set of tree oscillators with Bose–Hubbard and root-gauge interactions (Section 2, Section 3);
2. **Derives** the hydrodynamic/analogue-gravity reduction to an emergent curved metric (Section 5) and the Klein–Gordon to Schrödinger reduction (Section 6);

3. **Calibrates** its six parameters to six measured anchors with dimensionally consistent relations, exposing rather than hiding the hierarchy and cosmological-constant problems (Section 7);
4. **Separates** established results from conjectures (asymptotic safety, emergent Einstein dynamics, the sparticle spectrum) (Section 12);
5. **States** testable targets at the Planck, TeV, and cosmological scales (Section 8), flagged as model-dependent.

What the TOF provides is a concrete, internally consistent picture—analogue-gravity emergence of a curved metric from a discrete tree lattice, with the Standard Model sectors attached as a gauge/soliton structure—and a clear list of the computations that would be required to promote the conjectures to results.

References

- [1] Bombelli, L., Lee, J., Meyer, D., & Sorkin, R. D. (1987). Space-time as a causal set. *Phys. Rev. Lett.*, 59(5), 521.
- [2] Fisher, M. P. A., Weichman, P. B., Grinstein, G., & Fisher, D. S. (1989). Boson localization and the superfluid-insulator transition. *Phys. Rev. B*, 40(1), 546.
- [3] Unruh, W. G. (1981). Experimental black-hole evaporation? *Phys. Rev. Lett.*, 46(21), 1351.
- [4] Barceló, C., Liberati, S., & Visser, M. (2005). Analogue gravity. *Living Rev. Relativ.*, 8, 12.
- [5] 't Hooft, G. (1993). Dimensional reduction in quantum gravity. *arXiv: gr-qc/9310026*.
- [6] Rovelli, C. (2004). *Quantum Gravity*. Cambridge University Press.
- [7] Sorkin, R. D. (1995). A modified sum-over-histories for gravity. *arXiv: gr-qc/9507025*.
- [8] Weinberg, S. (1979). Phenomenological Lagrangians. *Physica A*, 96(1-2), 327.
- [9] Reuter, M. (1998). Nonperturbative evolution equation for quantum gravity. *Phys. Rev. D*, 57(2), 971.
- [10] Lindblad, G. (1976). On the generators of quantum dynamical semigroups. *Commun. Math. Phys.*, 48(2), 119.
- [11] Page, D. N. (1993). Information in black hole radiation. *Phys. Rev. Lett.*, 71(23), 3743.
- [12] Peebles, P. J. E., & Ratra, B. (2003). The cosmological constant and dark energy. *Rev. Mod. Phys.*, 75(2), 559.

A The Complete Lagrangian

For reference we collect the full Lagrangian with status labels.

$$\begin{aligned}
\mathcal{L}_{TOF} = & \underbrace{\frac{1}{16\pi G_{eff}} (R - 2\Lambda_{eff})}_{\text{Conjecture } G_1} - \underbrace{\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - \frac{1}{4} G_{\mu\nu}^a G^{a\mu\nu} - \frac{1}{4} W_{\mu\nu}^a W^{a\mu\nu}}_{\text{postulate}} \\
& + \underbrace{\sum_f \bar{\psi}_f (i\gamma^\mu D_\mu - m_f) \psi_f}_{\text{postulate}} + \underbrace{|D_\mu \phi|^2 - \left(-\frac{1}{2} \mu^2 |\phi|^2 + \frac{\lambda}{4!} |\phi|^4 \right)}_{\text{postulate}} \\
& + \underbrace{i\bar{\psi}_{SUSY} \not{\partial} \psi_{SUSY} + |\partial_\mu \phi_{SUSY}|^2 + \left(\frac{\partial W}{\partial \phi} \psi \psi + \text{h.c.} \right)}_{\text{optional, conjectural}} \\
& + \underbrace{\frac{1}{2} (\partial_\mu a)^2 - \Lambda_{QCD}^4 \left(1 - \cos \left(\frac{a}{f_a} \right) \right) - \frac{a}{f_a} G \tilde{G}}_{\text{postulate (soil axion)}} \\
& + \underbrace{\frac{1}{2} \int \prod_{i=1}^4 dg_i \phi(g_i) \phi(g_i) + \frac{\lambda_{GFT}}{5!} \int \prod_{i=1}^5 dg_i \phi(g_1) \cdots \phi(g_5)}_{\text{optional GFT tool}} \\
& + \underbrace{\mathcal{L}_{SK}[\phi_+, \phi_-, J_+, J_-]}_{\text{real-time tool}}.
\end{aligned} \tag{59}$$

B The TOF Equations of Motion

Assuming Conjecture 5.1 and the postulates of Section 3:

$$G_{\mu\nu} + \Lambda_{eff} g_{\mu\nu} = \frac{8\pi G_{eff}}{c^4} T_{\mu\nu}, \quad D_\mu F^{\mu\nu} = g J^\nu, \tag{60}$$

$$(i\gamma^\mu D_\mu - m_f) \psi_f = 0, \quad (\square + m_h^2) \phi = 0, \tag{61}$$

$$\square a + \frac{\partial V}{\partial a} = -\frac{1}{f_a} G \tilde{G}, \quad \frac{d\hat{\rho}}{dt} = -\frac{i}{\hbar} [\hat{H}, \hat{\rho}] + \sum_i \gamma_i \left(\hat{L}_i \hat{\rho} \hat{L}_i^\dagger - \frac{1}{2} \{ \hat{L}_i^\dagger \hat{L}_i, \hat{\rho} \} \right). \tag{62}$$

C The Holographic Principle

The Bousso bound is

$$S_{forest} \leq \frac{A}{4l_P^2}. \tag{63}$$

With the calibration $a = l_P$ (Eq. (33)), the maximum number of trees (bits) in a region is $N_{max} = A/a^2 = A/(4l_P^2)$, consistent with the bound. As in all holographic contexts, this is a counting statement about the fundamental lattice; it does not by itself explain *why* gravity holographically bounds entropy, which remains the deeper open problem.

D Age of the Forest

The Hubble time is

$$t_{forest} = \frac{1}{H_0} = \frac{1}{2.184 \times 10^{-18} \text{ s}^{-1}} = 4.58 \times 10^{17} \text{ s} \approx 14.5 \text{ Gyr.} \quad (64)$$

The measured age of the universe, 13.8 Gyr, is 0.95 times this, the standard correction for the matter content of the universe (Peebles & Ratra [12]).

End of Paper

The forest is the cosmos. The cosmos is the forest. The claims are separated from the conjectures, and the open problems are stated.
